ARSIPARIS

BLUEPRINT UTAMA PROJECT

MASTER BLUEPRINT — VERSI 6

Dokumen acuan utama untuk membangun aplikasi arsip digital secara bertahap, aman, teruji, dan realistis untuk level junior–mid.

Aspek	Keputusan / target
Stack	Laravel 13 + Inertia 3 + React 19 + TypeScript + Tailwind CSS 4 + MySQL
Arsitektur	Laravel backend + Inertia bridge + React/TypeScript UI; satu aplikasi
Core	Archive management, file, search, dashboard, audit, reporting
Intelligence	OCR → extraction/classification → confidence → human review → optional RAG
Security	Auth + authorization + private storage + validation + audit
Reliability	Queue + retry + idempotency + logging + backup/restore
Target	Aplikasi nyata yang selesai dan maintainable, bukan monster enterprise

Catatan versi: versi ini dibuat ulang sebagai master blueprint lengkap. Penomoran isi dimulai dari bagian 0 dan halaman PDF dimulai dari halaman 1; tidak ada bagian awal yang sengaja dipotong.

Basis pembaruan: Blueprint v4 yang menjadi dasar sebelumnya, ditambah detail spesifikasi database/ERD/DFD/workflow/AI/security/roadmap dan baseline repository/runtime yang telah dilaporkan selama pembangunan project.

0. Ringkasan Eksekutif

Arsiparis adalah sistem pengelolaan arsip digital yang mengelola dokumen dari upload sampai penyimpanan, pencarian, penggunaan, review, verifikasi, pelaporan, dan pengelolaan jangka panjang.

Arsitektur utama sengaja menggunakan **Laravel + Inertia + React + TypeScript**. Laravel tetap menjadi pusat routing, authentication, authorization, validation, business logic, dan akses database. React menjadi UI interaktif melalui Inertia tanpa membutuhkan frontend/API project terpisah.

Blueprint memisahkan **core system** dan **intelligence/advanced layer**. Core harus dapat digunakan ketika OCR/AI belum aktif atau sedang gagal. AI membantu pekerjaan, tetapi metadata final tetap berada di bawah kontrol manusia.

Target kualitas bukan kesempurnaan enterprise. Targetnya: selesai, konsisten, aman, dapat diuji, mudah dipahami, mudah dikembangkan, dan cukup matang sebagai portfolio project nyata.

0.1 Urutan besar

```
Foundation
→ Database
→ Auth & Authorization
→ Layout/Dashboard shell
→ Archive Core
→ File Management
→ Search
→ Dashboard
→ Audit
→ Reporting
→ OCR
→ AI
→ Queue/Reliability
→ Full-text/RAG (bila layak)
→ Testing/Security
→ Deployment/Backup/Monitoring
```

Prinsip utama: jangan membangun fitur advanced sebelum fondasi yang menjadi dependency-nya stabil.

1. Visi, Tujuan, Prinsip, dan Batasan

1.1 Tujuan

- Menyimpan arsip dan metadata secara terstruktur.
- Memudahkan pengguna menemukan dokumen melalui metadata, filter, dan pencarian.
- Menerapkan workflow upload → processing → review → verified → archived.
- Menerapkan role/permission dan policy di server.
- Mencatat aktivitas penting.
- Menyediakan reporting yang dapat diulang dan memiliki histori.
- Menambahkan OCR/AI secara bertahap tanpa membuat core bergantung padanya.

1.2 Prinsip desain

- **Core first:** fungsi utama tidak tergantung AI.
- **Human in the loop:** hasil OCR/AI adalah suggestion sampai diverifikasi.
- **Security by design:** authorization sebelum resource diberikan.
- **Progressive complexity:** Redis/search engine/vector DB/object storage tidak dipasang hanya karena bisa.
- **Simple but maintainable:** abstraksi muncul ketika kompleksitas nyata muncul.
- **Real workflow over demo:** state loading/empty/error/processing/failed/review harus nyata.

1.3 Batasan

- Tidak microservices.
- Tidak Next.js sebagai frontend kedua.
- Tidak API publik penuh tanpa kebutuhan.
- Tidak Kubernetes/event-driven/multi-tenancy enterprise tanpa kebutuhan.
- Tidak membiarkan AI mengambil keputusan final tanpa mekanisme verifikasi.

2. Baseline Project Saat Ini

Komponen	Baseline yang dilaporkan
Laravel	13.30.1
PHP	8.5.10
Composer	2.10.3
React	19.2.0
TypeScript	5.7.x
Inertia	3.x
Vite+	0.3.0; `npm run dev` memakai `vp dev`
Tailwind CSS	4.x
Database	MySQL — `arsiparis_db`
Laravel server	http://127.0.0.1:8000
Vite dev server	http://localhost:5173/
Session / Cache / Queue	database / database / database
Auth scaffolding	Fortify + passkey + 2FA tersedia
UI existing	Welcome, Login, Register dapat ditampilkan
Storage link	Belum linked pada baseline terakhir
Timezone	UTC pada `php artisan about`

2.1 Kondisi source tree yang sudah ada

```
app/
Actions/Fortify/
Concerns/
Http/Controllers/
Models/User.php
Providers/

database/
factories/
migrations/
seeders/

resources/js/
components/
hooks/
layouts/
pages/auth/
pages/dashboard.tsx
routes/
types/

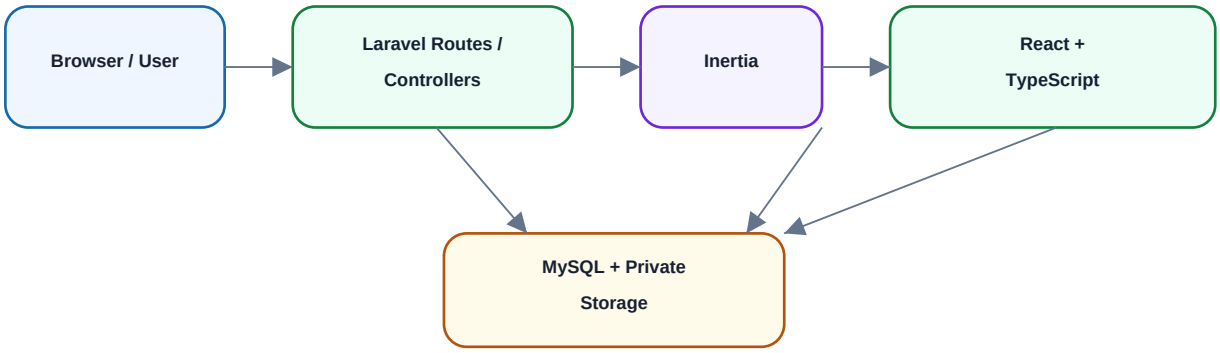
routes/
web.php
settings.php
console.php
```

2.2 Kesimpulan baseline

Fondasi framework dan authentication sudah tersedia. Karena itu, kita **tidak kembali mengulang setup stack**. Pekerjaan domain dimulai dari database/ERD core.

Warning optional package `fontaine` pada development bukan alasan untuk mengganti arsitektur.

3. Arsitektur Final



3.1 Tanggung jawab layer

Layer	Tanggung jawab
Routes	URL, HTTP method, middleware, route naming
Controller	Orkestrasi request → validation → authorization → domain action → response
Form Request	Validasi input dan authorization request bila sesuai
Policy	Hak akses terhadap resource/action
Model	Data, casts, relationships, query scopes sederhana
Service	Workflow/domain logic kompleks yang mulai berulang
Job	OCR/AI/report/process berat asynchronous
Inertia	Bridge server data ke React
React Page	Page composition dan UI state
Component	UI reusable
MySQL	Persistensi relational
Private Storage	File fisik arsip

3.2 Request flow standar

```
Browser
→ middleware/auth
→ Controller
→ Form Request validation
→ Policy authorization
→ Model/Service
→ DB/Storage
→ Inertia response
→ React
→ feedback state
```

4. Modul Sistem dan Scope

Modul	Fungsi	Prioritas
Authentication	Login/register/logout/reset/session/verification	CORE
User & Access	User, role, permission/policy	CORE
Archive Management	Metadata, master data, status, CRUD	CORE
File Management	Upload/private storage/preview/download	CORE
Search	Keyword/filter/sort/pagination	CORE
Dashboard	KPI/status/activity/trend	CORE
Audit	Activity/change tracking	CORE
Reporting	Daily/monthly/custom + PDF/Excel + history	INTERMEDIATE
OCR	Extract text dari scan/PDF	ADVANCED
AI	Metadata/classification/summary/confidence	ADVANCED
Queue	Background job/retry/failed/idempotency	ADVANCED
Full-text	Search isi OCR	ADVANCED
AI Assistant/RAG	Question answering berbasis arsip berizin	ADVANCED
Notification	Review/failure/approval alerts	ADVANCED
Versioning/Retention	Versi/trash/restore/retention	ADVANCED/OPEN

4.1 Fitur yang sengaja ditunda

- Real-time collaboration.
- Mobile native.
- Microservices.
- Public API platform.
- Vector DB sebelum RAG benar-benar diperlukan.
- Infrastructure enterprise yang belum punya alasan operasional.

5. Role, Permission, dan Authorization

Role	Tanggung jawab
Admin	User operasional, role/permission, master data, arsip, audit, approval, report
Staff	Upload, edit metadata, review OCR/AI, search, report yang diizinkan
Viewer	Search, view, download bila permission mengizinkan

5.1 Permission kandidat

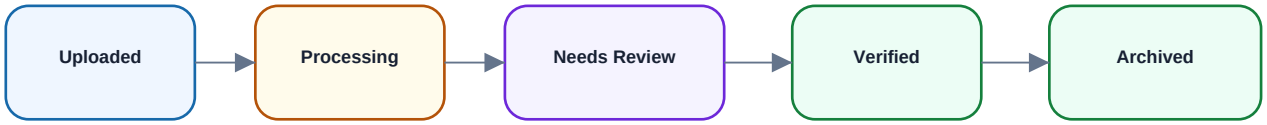
Permission	Arti
archive.view	Melihat arsip
archive.create	Membuat arsip
archive.update	Mengubah metadata
archive.delete	Soft/permanent delete sesuai policy
archive.approve	Verify/approve
file.download	Download file
report.generate	Generate report
user.manage	Kelola user
audit.view	Lihat audit
master.manage	Kelola master data

5.2 Aturan

- UI boleh menyembunyikan tombol, tetapi server tetap wajib mengecek Policy/permission.
- Download harus melalui endpoint terotorisasi.
- Jika akses nantinya berbasis unit/ownership, aturan tersebut ditempatkan pada policy.
- Perubahan permission dan tindakan sensitif masuk audit.

6. Workflow dan State Machine

Workflow inti: Upload → Validation → Processing → Needs Review → Verified → Archived. Failed adalah jalur pemulihan.



State	Makna	Transition berikutnya
Uploaded	Data/file awal diterima	Processing
Processing	OCR/AI/background job berjalan	Needs Review / Verified / Failed
Needs Review	Suggestion perlu diperiksa	Verified / kembali review
Verified	Metadata telah diperiksa	Archived
Archived	Arsip aktif	View/search/download
Failed	Processing gagal	Retry → Processing

6.1 Invariant

- AI tidak otomatis mengubah data Verified/Archived tanpa workflow yang eksplisit.
- Confidence membantu menentukan review, bukan menjamin kebenaran.
- Failure harus menjadi state yang terlihat, bukan timeout/blank page.
- State DB, UI, dan job harus konsisten.

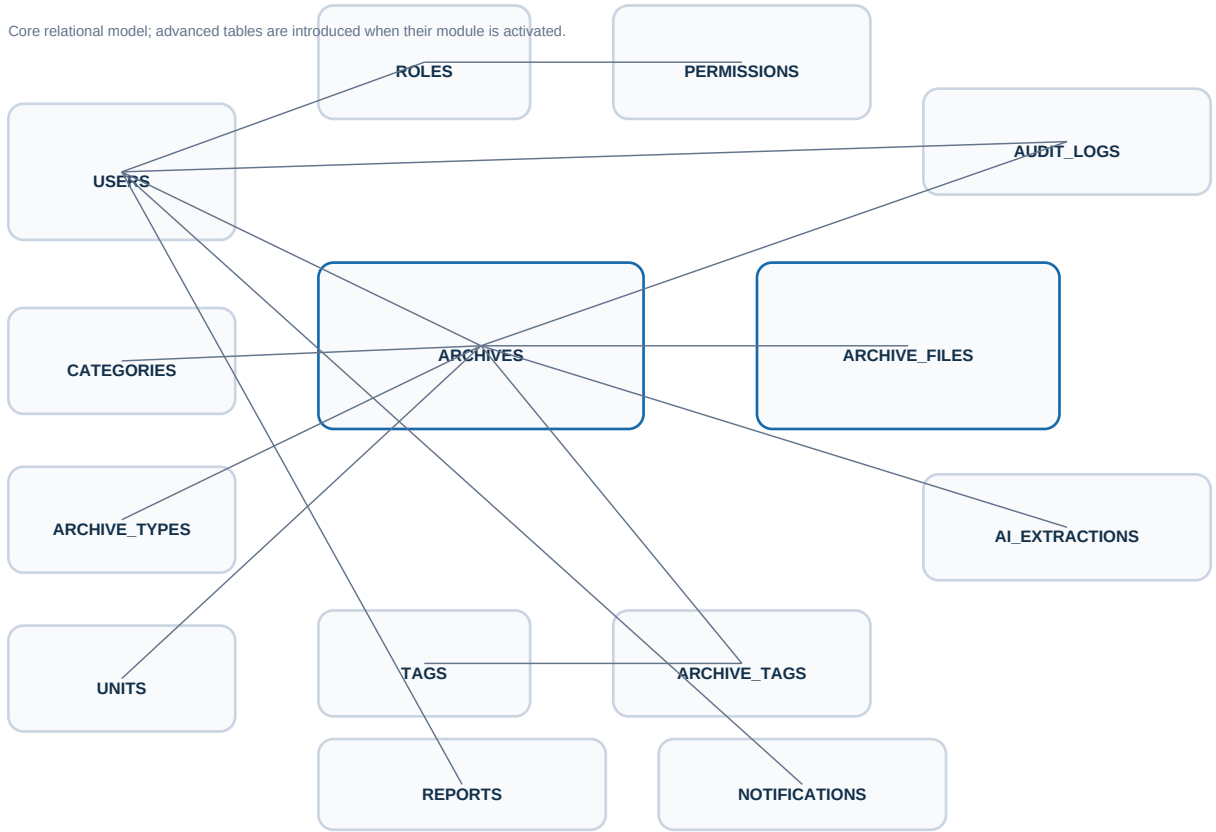
7. Database — Master Blueprint

Database menjadi fondasi sebelum UI domain kompleks. Tabel core dibangun dahulu; tabel advanced dapat ditambahkan ketika modulnya dimulai.

7.1 Kelompok tabel

Kelompok	Tabel	Tahap
Identity	users, roles, permissions, role_permissions	Core
Archive	archives, archive_files	Core
Master	categories, archive_types, units, tags, archive_tags	Core
Audit	audit_logs	Core/early
AI/OCR	ai_extractions	Advanced
Reporting	reports	Intermediate
Notification	notifications	Advanced
Versioning	archive_versions	Open/Advanced

7.2 ERD konseptual



Relasi utama: role 1:N users; role N:N permission; user 1:N archive; archive N:1 category/type/unit; archive 1:N file; archive N:N tag; archive/file 1:N AI extraction; user/archive 1:N audit; user 1:N reports/notifications.

8. Data Dictionary — Identity dan Archive Core

8.1 users

Kolom	Konseptual	Aturan
id	BIGINT	PK
name	VARCHAR	Required
email	VARCHAR	UNIQUE, required
email_verified_at	TIMESTAMP	Nullable
password	VARCHAR	Hashed; auth flow
role_id	FK roles	Required
status	ENUM/string	active/inactive/pending sesuai kebutuhan
remember_token	VARCHAR	Nullable
timestamps	TIMESTAMP	Laravel

8.2 roles / permissions / role_permissions

Tabel	Kolom inti	Constraint
roles	id, name, description	name UNIQUE
permissions	id, name, description	name UNIQUE
role_permissions	role_id, permission_id	UNIQUE pair + FKs

8.3 archives

Kolom	Aturan
id	PK
archive_number	Identifier; unique bila diwajibkan
document_date	Tanggal dokumen
subject	Perihal/judul
sender / recipient	Pengirim / penerima
archive_type_id / category_id / unit_id	FK master data
status	Workflow state
description	Opsional
uploaded_by	FK users
verified_by / verified_at	Nullable sampai verified
timestamps	created_at / updated_at
deleted_at	Optional soft delete

8.4 archive_files

Kolom	Aturan
id, archive_id	PK + FK
path	Private storage path
original_name	Metadata filename asli
stored_name	Nama internal aman/unik
mime_type / extension	Hasil validation
size	Bytes

Kolom	Aturan
sha256	Checksum; membantu duplicate/integrity
version	Default 1; naik jika versioning
uploaded_by	FK users
timestamps / deleted_at	Laravel + optional soft delete

9. Data Dictionary — Master, Audit, AI, Reporting

9.1 Master data

Tabel	Kolom inti	Catatan
categories	id, name, code?, description?, status	Reference category
archive_types	id, name, code?, description?, status	Jenis arsip
units	id, name, code?, description?, status	Unit/bagian
tags	id, name, slug?, timestamps	Tag; name UNIQUE
archive_tags	archive_id, tag_id	Pivot N:N; unique pair

9.2 audit_logs

Kolom	Fungsi
user_id	Actor
action	create/update/delete/verify/download/login/etc.
entity_type / entity_id	Resource yang dipengaruhi
before / after	JSON snapshot bila diperlukan
ip_address / user_agent	Context request
created_at	Waktu aktivitas

9.3 ai_extractions

Kolom	Fungsi
archive_id / archive_file_id	Target processing
provider / operation	Provider + OCR/extraction/classification/summary
input_hash / version	Identitas input
extracted_text	OCR text atau referensi
extracted_data	JSON field suggestions
classification	Suggested type/category/unit
confidence	Score dengan definisi konsisten
status	pending/processing/reviewed/accepted/rejected/failed
reviewed_by / reviewed_at	Human review
error_code / message	Safe failure detail

9.4 reports / notifications / archive_versions

Tabel	Kolom inti	Tujuan
reports	created_by, type, parameters, format, output_path, status, timestamps	Histori + reproducibility
notifications	user_id, type, data, read_at, timestamps	Review/failure/approval alert
archive_versions	archive_id, version, metadata_snapshot, file_id?, created_by	Version history bila diperlukan

10. Database Rules, Index, Transaction, Migration

10.1 Constraint

- Primary key seluruh tabel.
- Foreign key pada relasi penting.
- Unique pada email, role name, permission name, tag name, dan identifier arsip sesuai policy.
- Unique composite pada role_permissions dan archive_tags.
- NOT NULL hanya untuk data yang benar-benar wajib.

10.2 Index awal

Tabel	Index kandidat	Alasan
users	email UNIQUE, role_id, status	Login/filter
archives	archive_number, document_date, status, category_id, archive_type_id, unit_id, uploaded_by	Search/filter/join
archive_files	archive_id, sha256, created_at	File lookup/integrity
audit_logs	user_id, entity_type+entity_id, created_at, action	Activity query
ai_extractions	archive_file_id, status, created_at	Processing/review queue
reports	created_by, type, status, created_at	History
notifications	user_id, read_at, created_at	Unread list

Rule: jangan membuat index sebanyak mungkin. Tambahkan berdasarkan query nyata dan pengujian.

10.3 Transaction boundaries

- Archive + initial file record bila harus atomik.
- Verify + audit event.
- Role/permission changes.
- Job result + processing state.

10.4 Migration order

```
users/auth baseline
→ roles
→ permissions
→ role_permissions
→ categories/types/units/tags
→ archives
→ archive_files
→ archive_tags
→ audit_logs
→ reports
→ ai_extractions
→ notifications / archive_versions (when needed)
```

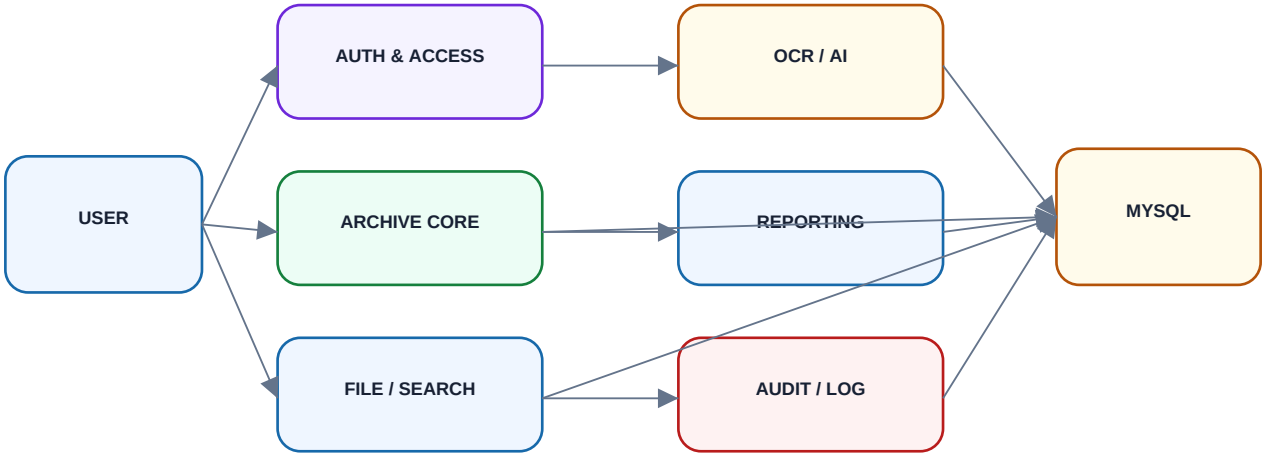
11. DFD — Data Flow Diagram

11.1 Context / Level 0



Context diagram memandang Arsiparis sebagai satu sistem yang menerima request/data dan mengembalikan hasil/status. OCR/AI adalah service layer di dalam workflow, bukan aplikasi utama yang terpisah.

11.2 Level 1



11.3 Level 2 — Upload → OCR → AI → Review

Proses	Input	Output
Receive upload	File + metadata	Validated request
Validate/store	Validated file	Private file + archive
Queue	Archive/file ID	Processing job
OCR	File	Extracted text
AI extraction	Text + context	Suggested metadata + confidence
Classification	Text + master data	Suggested class
Review	Suggestion + source	Corrected values
Verify/index	Reviewed metadata	Verified archive + searchable data

12. Sequence dan Use Case Utama

12.1 Login

```
Browser → POST /login → Laravel Auth
→ verify user → create/update session
→ Inertia redirect → React dashboard
```

12.2 Upload archive

```
Browser
→ Controller
→ validate + authorize
→ DB archive + private file
→ dispatch processing job
→ return Processing state
→ job: OCR/AI
→ persist result
→ update state
→ UI reads current status
```

12.3 Human review

```
Reviewer opens Review Queue
→ Laravel loads archive + AI/OCR result
→ React shows source + suggestions
→ Reviewer edits/accepts
→ Laravel validates + authorizes
→ save verified metadata
→ audit event
→ status Verified
→ archive becomes searchable
```

12.4 Download

```
GET /archives/{archive}/files/{file}/download
→ auth
→ policy authorization
→ file existence check
→ stream private storage
→ audit download
```

12.5 Use case matrix

Aktor	Use case
Admin	Users, roles, master data, archives, audit, reports, approval
Staff	Create/upload/edit/review/search/report
Viewer	Search/view/download if permitted
Job worker	OCR, AI, report, retry, state transition

13. File Management dan Storage Security

13.1 Upload pipeline

```
File
→ extension + MIME + size validation
→ authorization
→ private storage
→ archive_files record
→ hash/checksum if appropriate
→ processing trigger
→ UI status
```

13.2 Security rules

- Jangan percaya extension dari filename saja.
- Validasi extension, MIME, dan ukuran.
- Stored filename harus aman/unik.
- Original filename hanya metadata.
- File sensitif berada di private storage.
- Download hanya melalui endpoint terotorisasi.
- Preview juga harus melewati authorization.
- Hash dapat membantu duplicate detection/integrity.

13.3 Soft delete / trash

- Tambahkan setelah CRUD stabil.
- Restore membutuhkan policy.
- Permanent delete membutuhkan permission lebih tinggi dan audit.

13.4 Fallback

Jika preview tidak aman atau format tidak didukung, aplikasi tetap menyediakan metadata dan download terotorisasi. Jangan mengorbankan security demi preview.

14. Search, Dashboard, dan Reporting

14.1 Search core

Fitur	Target
Keyword	archive_number, subject, sender, recipient, tags
Filter	category, type, unit, year, status, date range
Sorting	newest, oldest, archive number
Pagination	Server-side
Empty state	Jelaskan hasil kosong + action

Full-text isi dokumen ditambahkan setelah OCR menghasilkan text yang konsisten. Search engine khusus tetap optional.

14.2 Dashboard

Widget	Nilai
Total Arsip	Jumlah aktif
Incoming/Outgoing	Distribusi jenis
Pending Review	Dokumen perlu tindakan
Processing/Failed	Health processing
Archive per Month	Trend
Recent Activity	Upload/edit/verify/download/delete

14.3 Reporting

```
Pilih report type
→ periode
→ filter
→ preview/ringkasan
→ generate
→ simpan history
→ download PDF/Excel
```

- Daily, monthly, dan custom report.
- History menyimpan creator, time, parameters, format, status, output.
- Template dapat berisi logo/header, judul, ringkasan, tabel, statistik, footer.

15. OCR dan AI — Intelligence Layer

AI adalah nilai tambah penting, tetapi ditempatkan setelah core system. Jika AI provider mati, core archive/search berbasis metadata tetap bekerja.

15.1 Capability map

Capability	Input	Output
OCR	Scan/PDF/image	Extracted text
Metadata extraction	Text + context	Number/date/subject/sender/recipient suggestions
Classification	Text + master data	Type/category/unit suggestion
Summary	Text	Short summary
Natural-language search	User query	Search intent/filter assistance
RAG assistant	Authorized indexed text	Answer + source documents

15.2 Pipeline

```
Document
→ OCR
→ Extracted Text
→ AI Extraction / Classification
→ Confidence
→ Human Review
→ Verified Metadata
→ Search / Report
→ optional Retrieval / RAG
```

15.3 Human review policy

- AI output adalah suggestion.
- Reviewer dapat mengedit.
- Low confidence dapat otomatis masuk Needs Review.
- Hasil accepted/edited disimpan untuk audit.
- Verified metadata menjadi source of truth.

15.4 Confidence

Threshold final tidak ditentukan secara asal. Kita uji pada sample dokumen nyata. Confidence dipakai sebagai sinyal risiko/review, bukan bukti kebenaran.

16. Natural-Language Search dan RAG Assistant

16.1 Konsep RAG

```
Documents
→ OCR/text
→ chunking
→ indexing
→ retrieval
→ authorization filter
→ model
→ answer + source documents
```

16.2 Contoh pertanyaan

- “Cari surat masuk dari Dinas Pendidikan tentang kegiatan tahun 2025.”
- “Apa isi utama surat nomor 045/ABC/2025?”
- “Berapa dokumen terkait kegiatan X pada tahun 2025?”

16.3 Security invariant

Retrieval harus authorization-aware. Model tidak boleh menerima chunk dari dokumen yang user sebenarnya tidak boleh lihat. Filter permission dilakukan sebelum context diberikan ke model.

16.4 Kapan RAG layak?

- OCR text sudah cukup konsisten.
- Metadata dan permission sudah stabil.
- Search core sudah berguna.
- Ada use case pencarian bahasa alami yang nyata.
- Biaya/provider/privacy sudah dapat diterima.

17. Queue, Reliability, Idempotency, Backup

17.1 Job lifecycle

```
Pending → Processing → Success → Persist result → Update state

Error:
Processing → Retry (bounded) → Failed → Manual retry/fallback
```

17.2 Reliability minimum

- Retry dengan batas.
- Timeout.
- Failed jobs dapat diperiksa.
- Logging error.
- Idempotency.
- Processing state terlihat di UI.

17.3 Idempotency model

```
identity =
archive_file_id
+ operation
+ input_hash
+ provider/model

Jika identity sudah sukses:
→ jangan membuat hasil duplikat.

Jika processing:
→ jangan enqueue pekerjaan identik kedua.
```

17.4 Backup

- Backup database.
- Backup private archive files.
- Dokumentasikan restore procedure.
- Sebelum production, lakukan restore test.

Backup yang tidak pernah diuji restore belum dapat dianggap reliable.

18. Audit, Notification, Versioning, Retention

18.1 Audit events

Event	Catat?	Contoh
Login/logout	Ya	Authentication event
Upload	Ya	Archive/file created
Metadata update	Ya	Subject/category changed
Verify	Ya	Reviewer + timestamp
Download	Ya	Sensitive file access
Delete/restore	Ya	Lifecycle change
Permission change	Ya	Role/permission changed
AI review	Ya	Accepted/rejected suggestion
Report generation	Ya	Report created

18.2 Notification

- Pending review, processing failed, approval, report complete.
- Mulai dari database notification/polling; realtime hanya bila dibutuhkan.

18.3 Versioning

- Tidak wajib pada fase awal.
- Jika diperlukan, archive_versions menyimpan snapshot metadata + file reference.
- Version number harus konsisten per archive.

18.4 Retention

- Jangan membuat retention policy kompleks sebelum kebutuhan bisnis/legal jelas.
- Trash/soft delete berbeda dari retention.

19. UI/UX Information Architecture

Area	Halaman
Public	Welcome
Auth	Login, Register, Forgot/Reset, Verification, 2FA/Passkey
Dashboard	KPI, processing, review, activity
Archives	List, Create, Detail, Edit
Review	Queue, Review Detail
Master	Categories, Types, Units, Tags
Users	List, Detail, Role
Audit	Activity Log
Reports	Generator, History
Settings	Profile, Security, Appearance, needed app settings

19.1 Archive Detail

- Archive number + status.
- Tanggal, subject, sender, recipient.
- Type/category/unit/tags.
- Description.
- File list + metadata + version bila ada.
- Preview bila aman.
- OCR text.
- AI suggestion + confidence + review state.
- Activity/history.
- Actions sesuai permission.

19.2 UI state quality bar

- Loading, empty, success, error, processing, failed, needs review.
- Destructive action memiliki confirmation.
- Responsive dan usable.
- Table/list menjadi pusat kerja, bukan dashboard penuh grafik.

20. Struktur Kode dan Development Practice

20.1 Backend

```
app/
■■ Http/Controllers/
■■ Http/Requests/
■■ Models/
■■ Policies/
■■ Services/ # hanya bila logic kompleks
■■ Jobs/ # OCR/AI/report
■■ Providers/
■■ Concerns/

database/
■■ migrations/
■■ factories/
■■ seeders/

routes/
■■ web.php
■■ settings.php
■■ console.php
```

20.2 Frontend

```
resources/js/
■■ pages/
■ ■■ auth/
■ ■■ archives/
■ ■■ review/
■ ■■ reports/
■ ■■ users/
■ ■■ master/
■ ■■ settings/
■■ components/
■■ layouts/
■■ hooks/
■■ types/
■■ lib/
■■ actions/
■■ routes/
■■ wayfinder/
```

20.3 Rules

- Tidak membuat repository/service untuk setiap CRUD sederhana.
- Extract reusable component saat pola benar-benar berulang.
- Gunakan TypeScript pada props/data penting.
- Form Request untuk validation kompleks.
- Policy untuk authorization.
- Transaction untuk operasi multi-write atomik.
- Commit Git berdasarkan unit kerja.

21. Testing, Security Hardening, Quality Bar

21.1 Testing

Level	Contoh
Unit	State transition, confidence policy, helper/service
Feature	CRUD, upload, search, verify, permission
Integration	Storage + DB + queue/job
Browser/E2E	Login → create → upload → review → verify → download

21.2 Test wajib

- Authentication + authorization.
- CRUD archive.
- Upload validation.
- Unauthorized download.
- Search/filter/pagination.
- Review → verify.
- Soft delete/restore bila aktif.
- OCR/AI review flow ketika aktif.
- MIME mismatch, oversized file, duplicate job, processing failure.

21.3 Security minimum

Area	Minimum
Auth	Hashing, secure session, reset, rate limiting if needed
Authorization	Policy/permission pada archive/file/user/report/audit
Upload	Extension + MIME + size + safe filename
Storage	Private + authorized download
Validation	Form Request + business rules
Audit	Sensitive actions recorded
Secrets	No API keys/password in source control
AI	Provider key server-side; retrieval respects authorization

22. Deployment, Backup, Monitoring

22.1 Production checklist

- APP_ENV=production.
- APP_DEBUG=false.
- Secrets aman; tidak di repository.
- HTTPS/SSL aktif.
- Storage permission benar.
- DB credentials aman.
- Queue worker aktif bila processing async.
- Scheduler aktif bila diperlukan.
- Logs/error monitoring tersedia.

22.2 Backup scope

Objek	Backup / recovery
MySQL	Full DB + tested restore
Private files	Archive files + mapping metadata
Reports	Jika output report merupakan data penting
Configuration	Recovery procedure; secrets dikelola aman

22.3 Monitoring minimum

- Application errors.
- Queue failures.
- Processing duration.
- Storage/database health.
- Failed OCR/AI count.
- Authentication anomalies bila relevan.

23. Roadmap Implementasi Utama

Roadmap bukan jadwal kaku. Tahap berikutnya dimulai ketika tahap sebelumnya cukup stabil.

#	Tahap	Output
01	Foundation	Environment, Git, base layout, project baseline
02	Database	ERD, migrations, models, relationships, seed
03	Auth & Access	Role, permission, policy, user status
04	Archive Core	CRUD, metadata, master data, workflow
05	File Management	Validation, private storage, preview/download
06	Search	Keyword/filter/sort/pagination
07	Dashboard	KPI, review/processing, recent activity
08	Audit	Activity log, soft delete/restore
09	Reporting	Daily/monthly/custom, PDF/Excel, history
10	OCR	Text extraction + status + review
11	AI	Extraction/classification/confidence/review
12	Queue	Background jobs/retry/failed/idempotency
13	Advanced Search	Full-text; semantic if justified
14	AI Assistant	RAG + authorization-aware retrieval
15	Advanced Ops	Notification/versioning/retention if needed
16	Quality	Testing/security/performance/UX
17	Deployment	Server/cloud/SSL/backup/monitoring/docs

23.1 Milestone gates

- Gate 1: database dapat fresh migrate + seed.
- Gate 2: auth + role/policy benar.
- Gate 3: archive CRUD + workflow benar.
- Gate 4: private file + authorization test lulus.
- Gate 5: search/dashboard/audit stabil.
- Gate 6: reporting stabil.
- Gate 7: OCR/AI prototype teruji sebelum productionizing.

24. Keputusan Locked/Open dan Risiko

24.1 Decision register

Keputusan	Status	Guardrail
Laravel + Inertia + React	LOCKED	Tidak pindah tanpa alasan kuat
React + TypeScript	LOCKED	Tidak tambah frontend framework kedua
MySQL	LOCKED	Search engine hanya bila terbukti perlu
Private storage	LOCKED	No public direct access
Human review AI	LOCKED	AI ≠ source of truth
Queue	PLANNED	Aktif saat processing berat
AI/OCR provider	OPEN	Benchmark akurasi/biaya/privacy
Vector DB	OPEN	Hanya untuk RAG yang nyata
Full versioning	OPEN	Tunggu kebutuhan histori
Realtime notification	OPEN	Mulai sederhana

24.2 Risiko dan mitigasi

Risiko	Mitigasi
Scope terlalu besar	Roadmap + milestone gate
AI terlalu cepat	Core DB/workflow/permission/file dulu
AI dianggap benar	Confidence + human review + audit
File bocor	Private storage + authorized endpoint
UI hanya screenshot	Workflow + state wajib
Over-engineering	Tambah abstraction saat kompleksitas nyata
Request lambat	Queue untuk proses berat
Search buruk	Metadata/filter dari awal; full-text setelah OCR
Backup tidak pulih	Restore test

25. Langkah Kita Sekarang — Eksekusi Konkret

Kondisi project saat ini sudah memiliki Laravel/Inertia/React dan authentication baseline. Karena itu langkah berikutnya adalah **database core**, bukan mengganti stack atau mengerjakan AI.

Urutan	Pekerjaan	Definition of done
1	Finalisasi ERD core	Relasi identity/master/archive/file jelas
2	Migrations	Schema dapat dibangun dari DB kosong
3	migrate fresh	Tidak ada migration error
4	Models + relations	Eloquent relation dapat diuji
5	Factories/seeders	Role/master data tersedia
6	Development users	Admin/Staff/Viewer tersedia
7	Relation tests	Archive/user/file/master query benar
8	Git commit	Database foundation terisolasi

25.1 Perintah development yang relevan

```
php artisan migrate:fresh --seed
php artisan db:show
php artisan route:list
php artisan about
npm run dev
npm run types:check
npm run check
```

Perhatian: `migrate:fresh` menghapus seluruh tabel pada database target. Gunakan hanya pada database development yang datanya memang boleh di-reset.

25.2 Yang belum kita kerjakan sekarang

- Belum memasang provider AI.
- Belum membuat RAG/vector DB.
- Belum membuat search engine khusus.
- Belum membuat infrastructure enterprise.
- Belum memaksimalkan dashboard sebelum data core tersedia.

26. Definition of Done dan Master Checklist

26.1 Sebuah modul dianggap selesai jika

- Happy path end-to-end berjalan.
- Validation tersedia.
- Authorization sesuai role.
- Error utama ditangani.
- Loading/empty/success state tersedia.
- Data dan relasi benar.
- Tidak ada file/data leak.
- Test kritis tersedia.
- Kode dapat dipahami tanpa abstraksi tidak perlu.
- Perubahan tidak merusak modul sebelumnya.

26.2 Final release

- Production debug off.
- Secrets aman.
- HTTPS.
- DB + file backup.
- Restore pernah diuji.
- Queue/scheduler bila diperlukan.
- Logging/monitoring.
- Installation/deployment documentation.

26.3 Master principle

Jika muncul ide baru selama coding, ide tersebut tidak otomatis masuk scope. Kita cek: masalah apa yang diselesaikan, apakah dibutuhkan sekarang, apakah manfaat sebanding dengan kompleksitas, dan apakah memengaruhi keputusan locked? Jika memengaruhi arsitektur/schema/workflow/security, blueprint diperbarui sebelum keputusan dianggap final.

Kesimpulan: Arsiparis dibangun sebagai aplikasi arsip yang benar-benar bekerja terlebih dahulu, lalu berkembang menuju OCR, AI, search cerdas, reliability, dan RAG secara bertahap. Core tetap bernilai walaupun intelligence layer belum aktif.

Lampiran A. Checklist ERD Sebelum Migration

- Semua tabel memiliki PK.
- Semua FK penting sudah jelas.
- Relasi N:N menggunakan pivot.
- Nullable field memang opsional.
- Unique constraint sudah dipertimbangkan.
- Index didasarkan pada query nyata.
- Status memiliki daftar nilai yang jelas.
- Audit requirement sudah diketahui.
- Tabel advanced tidak dipaksakan masuk migration awal.

A.1 Archive schema reference

```
archives
id
archive_number
document_date
subject
sender
recipient
archive_type_id
category_id
unit_id
status
description
uploaded_by
verified_by
verified_at
created_at
updated_at
deleted_at (optional)

archive_files
id
archive_id
path
original_name
stored_name
mime_type
extension
size
sha256
version
uploaded_by
created_at
updated_at
deleted_at (optional)
```

Lampiran B. Checklist File Security

- Upload hanya oleh user berhak.
- Extension + MIME + size tervalidasi.
- Stored filename aman/unik.
- Original filename hanya metadata.
- Private storage digunakan.
- Download memakai policy.
- Preview tidak bypass authorization.
- Unauthorized URL access test tersedia.
- Error tidak membocorkan filesystem path/secrets.
- Download tercatat bila dibutuhkan.

Lampiran C. Checklist OCR/AI

- Core archive CRUD stabil.
- Role/policy stabil.
- Private storage stabil.
- Queue tersedia saat processing berat.
- Sample dokumen nyata sudah diuji.
- Provider dinilai berdasarkan akurasi, biaya, privacy, latency, stability.
- Confidence policy diuji.
- Low confidence masuk review.
- Verified metadata menjadi source of truth.
- Failure menjadi Failed state.
- Provider key hanya di server.
- RAG retrieval menerapkan authorization.
- Assistant memberikan sumber dokumen bila memungkinkan.

Lampiran D. Glossary

Istilah	Arti
Archive	Entitas utama dokumen + metadata.
Archive File	File fisik terkait archive.
Master Data	Category/type/unit/tag.
Policy	Aturan authorization Laravel.
Workflow	State dan transition arsip.
OCR	Mengubah scan/image menjadi text.
Extraction	Mengambil field metadata dari text.
Classification	Saran category/type/unit.
Confidence	Sinyal tingkat keyakinan model.
Human-in-the-loop	Manusia memeriksa hasil otomasi.
RAG	Retrieval-Augmented Generation.
Idempotency	Retry tidak menghasilkan efek duplikat tak diinginkan.
Audit Log	Jejak aktivitas/perubahan penting.
Soft Delete	Tandai deleted tanpa langsung menghapus row.

STATUS: MASTER BLUEPRINT V6. Dokumen ini adalah acuan utama kerja. Detail provider, threshold, infrastructure, dan beberapa tabel advanced dapat berubah setelah prototype/testing, tetapi perubahan material harus dicatat sebagai keputusan baru dan tidak dilakukan diam-diam.